

Binary Numbers

Task 1: a) $4310_5 = 4 \cdot 5^3 + 3 \cdot 5^2 + 1 \cdot 5^1 + 0 \cdot 5^0$
 $= 500 + 75 + 5$
 $= \underline{\underline{580}}$

b) $198_{12} = 1 \cdot 12^2 + 9 \cdot 12^1 + 8 \cdot 12^0$
 $= 144 + 108 + 8$
 $= \underline{\underline{260}}$

Task 2:

$\frac{27}{2} = 13$	1	$0.315 \cdot 2 = 0.630 = 0$
$\frac{13}{2} = 6$	1	$0.630 \cdot 2 = 1.260 = 1 \text{ (.260 rest)}$
$\frac{6}{2} = 3$	0	$0.260 \cdot 2 = 0.520 = 0$
$\frac{3}{2} = 1$	1	$0.520 \cdot 2 = 1.040 = 1 \text{ (.040 rest)}$
$\frac{1}{2} = 0$	1	$0.040 \cdot 2 = 0.080 = 0$

11011.01010

Task 3: a) $\overset{M}{10011} - \overset{N}{10010}$

$2's \text{ comp}(N) = 01101 + 1$	$M + 0110$	Answer
$= 01110$	10011	<u><u>00001</u></u>
	$+ 01110$	
	$= 100001$	

Task 3: b) $100010^M - 100110^N$

	$M + 011010$	$2's \text{ comp because negative}$
$2's \text{ comp}(N) = 011001 +$	100010	$000011 + 1$
$= 011010$	$+ 011010$	$= \underline{\underline{000100}}$
	$= 111100$	

c) $1001 - 110101$, M need more bits
 $0001001 - 0110101$

	$M + 1001011$	$2's$
$2's = 1001010 + 1$	0001001	$0101011 + 1$
$= 1001011$	$+ 1001011$	$= \underline{\underline{0101100}}$
	$= 1010100$	

d) $101000 - 10101$
 $101000 - 010101$

	$M + 101011$	$Answer$
$2's = 101010 + 1$	101000	$\underline{\underline{011011}}$
$= 101011$	$+ 101011$	
	$= 1011011$	

Task 4: a) 7 bits

b) $29 - 49$, $0011101 (+29) - 0110001 (+49)$
 $2's = 1001110 + 1 = 1001111 \Rightarrow (-49)$

$M + N$	$2's \text{ and convert to decimal}$
0011101	$0010011 + 1 = 0010100 = 20$
$+ 1001111$	
$= 1101100$	$\underline{\underline{-20}}$

Task 4: c) $-29 + 49$, $001101 (+29) - 0110001 (+49)$
 $2's = 1100010 + 1 = 1100011 = 7(-29)$

$ \begin{array}{r} m+n \\ 1100011 \\ + 0110001 \\ \hline = 10010100 \end{array} $	$0010100 = \underline{\underline{+20}}$
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d) $-29 - 49$, $001101 (+29) - 0110001 (+49)$
 $2's = 1100010 + 1 = 1100011 = 7(-29)$
 $2's = 1001110 + 1 = 1001111 = 7(-49)$

Add 1 in front because you need 8 bits for it to be mathematically correct

$ \begin{array}{r} m+n \\ 11100011 \\ + 11001111 \\ \hline = 10110010 \end{array} $	$ \begin{array}{l} 2's \text{ and convert to decimal} \\ 01001101 + 1 = 01001110 = 78 \\ \underline{\underline{-78}} \end{array} $
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Task 5: a) 6, 3, 1, 1

$0 = 0, 0, 0, 0$
 $1 = 0, 0, 0, 1$
 $2 = 0, 0, 1, 1$
 $3 = 0, 1, 0, 0$
 $4 = 0, 1, 1, 0$ or $0, 1, 0, 1$
 $5 = 0, 1, 1, 1$
 $6 = 1, 0, 0, 0$
 $7 = 1, 0, 1, 0$ or $1, 0, 0, 1$
 $8 = 1, 0, 1, 1$
 $9 = 1, 1, 0, 0$

b) 6, 4, 2, 1

$0 = 0, 0, 0, 0$
 $1 = 0, 0, 0, 1$
 $2 = 0, 0, 1, 0$
 $3 = 0, 0, 1, 1$
 $4 = 0, 1, 0, 0$
 $5 = 0, 1, 0, 1$
 $6 = 1, 0, 0, 0$ or $0, 1, 1, 0$
 $7 = 1, 0, 0, 1$ or $0, 1, 1, 1$
 $8 = 1, 0, 1, 0$
 $9 = 1, 0, 1, 1$